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| Zewail City School of Science and Technology | Group: $4^{th}$ Year |
| School of Computational Sciences and | Course Name: Speech Recognition |
| Artificial Intelligence | Course Code: DSAI 456 |
| Date: November 13, 2025 | Time Allowed: 1.25 hours |
| Total Marks: 20 | No. of Pages: 6 |

Answer the questions in the spaces provided on the question sheets.

## GROUP A

Name: _____

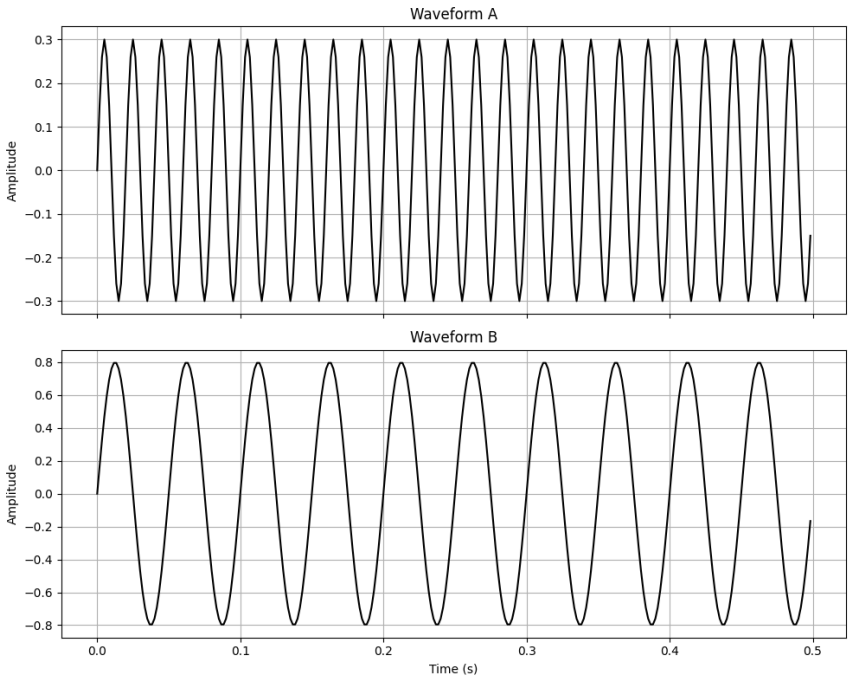
Serial No.: _____

| Question | Points | Score |
|---|---|---|
| 1 | 5 | |
| 2 | 2 | |
| 3 | 1 | |
| 4 | 4 | |
| 5 | 4 | |
| 6 | 4 | |
| Total: | 20 | |

In **ALL** questions, you need to show **all the steps** required to obtain your answers.

Do not answer in this page!

**Q1** Given the following waveforms




(a) (1 point) What is the frequency of waveform B?

Answer:

(b) (1 point) What is the intensity of waveform A?

Answer:

(c) (1 point) Which waveform corresponds to a louder speech?

Answer:

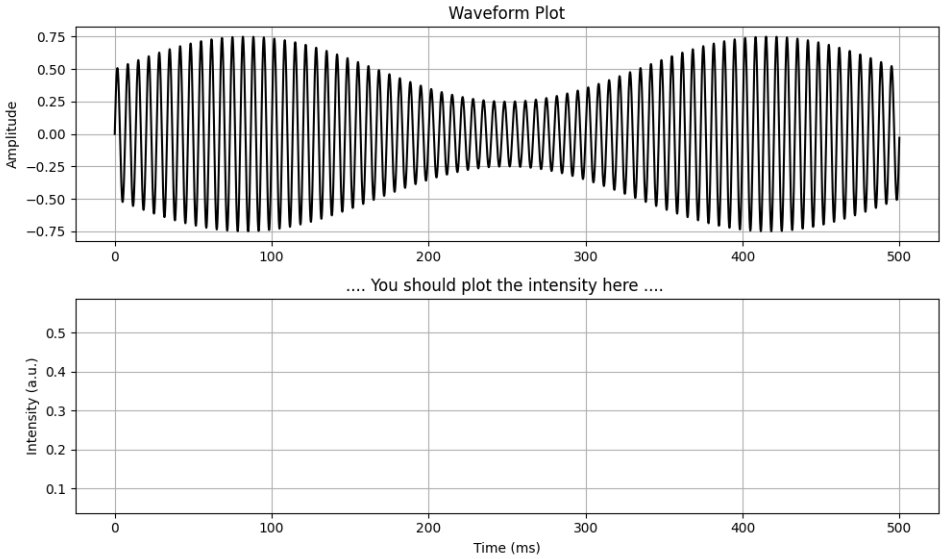
(d) (1 point) Which waveform corresponds to a speech with a higher pitch?

Answer:

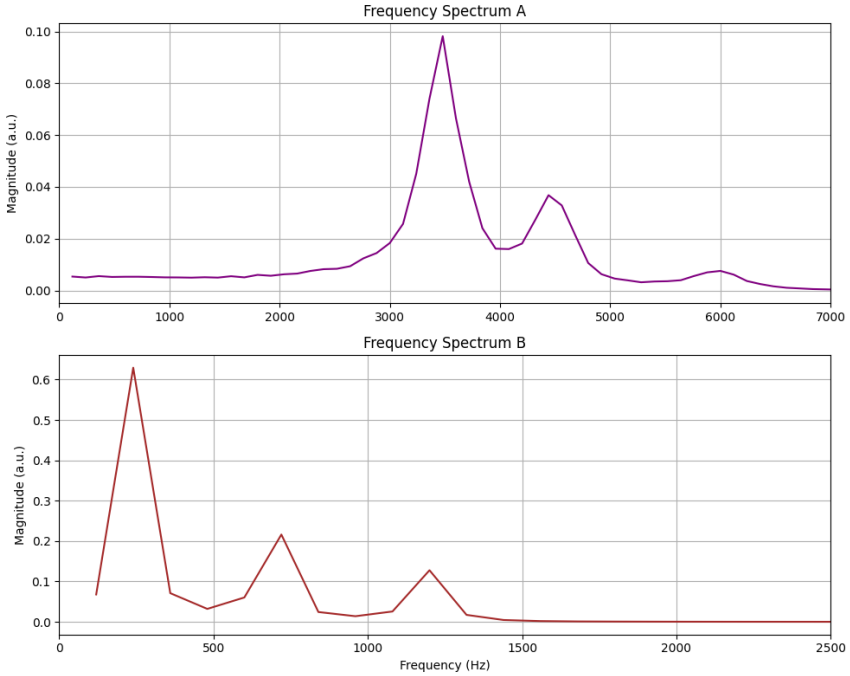
(e) (1 point) Can a signal have high pitch and low loudness simultaneously? Explain how.

Answer:

**Q2** (2 points) Plot the intensity of the following waveform in the empty plot



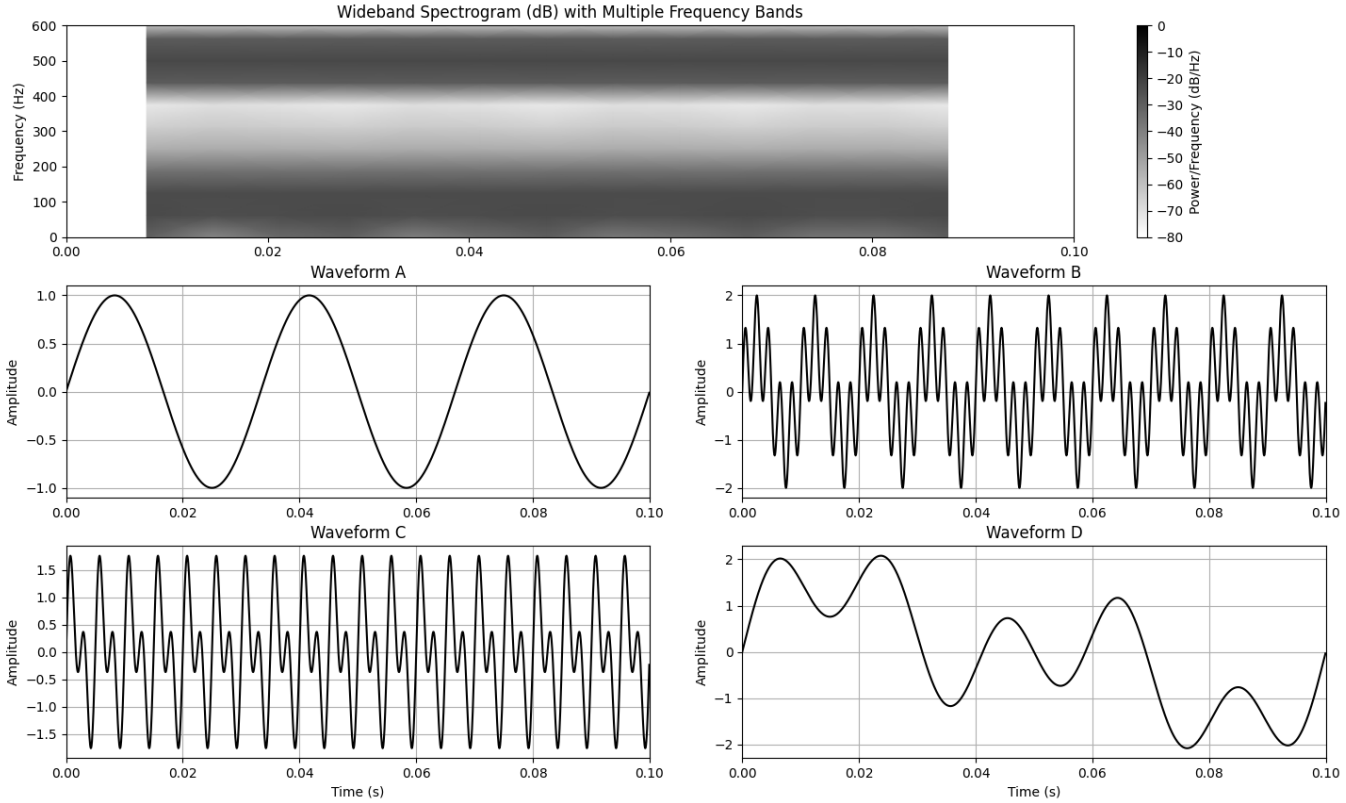

**Q3** The fricative phone [s] is characterized by high-frequency noisy formants while the voiced nasal fricative [m] has lower frequency formants. Given the following two frequency spectrum.




(a) (1 point) Which one represents the spectrum for phone [m]? Justify.

Answer:

Q4 The following figure displays a spectrogram in the upper panel and four waveforms arranged below it.




(a) (2 points) Identify and explain the visible frequency bands in the spectrogram.

Answer:

(b) (2 points) Which waveform corresponds to the spectrogram? Explain why.

Answer:

Q5 Consider a speech recognition system designed to recognize two spoken digits: *two* and *three*. The hidden states are the digits themselves $S = \{\text{two}, \text{three}\}$. The observations are the acoustic feature categories $O = \{\text{low energy}, \text{medium energy}, \text{high energy}\}$. The HMM parameter set $\lambda = (A, B, \pi)$ where:

- $\pi = \begin{bmatrix} 0.1 & 0.1 \end{bmatrix}$
- $A = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$
- $B = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.5 & 0.3 & 0.2 \end{bmatrix}$

(a) (4 points) Given the observed acoustic feature sequence $O = [\text{medium energy}, \text{high energy}, \text{high energy}]$, calculate the likelihood $P(O|\lambda)$ using the **Forward** algorithm.

Q6 For Gaussian Mixture Models (GMM):

(a) (1 point) Write the objective function that the Expectation-Maximization (EM) algorithm seeks to maximize when estimating the GMM parameters.

(b) (2 points) Outline the E-step and M-step equations used in the EM algorithm for GMM parameter estimation, explaining how responsibility values for each Gaussian component are computed and used.

(c) (1 point) Explain why covariance matrices are often assumed to be diagonal in speech recognition tasks, and discuss the implications of this assumption on model complexity and performance.